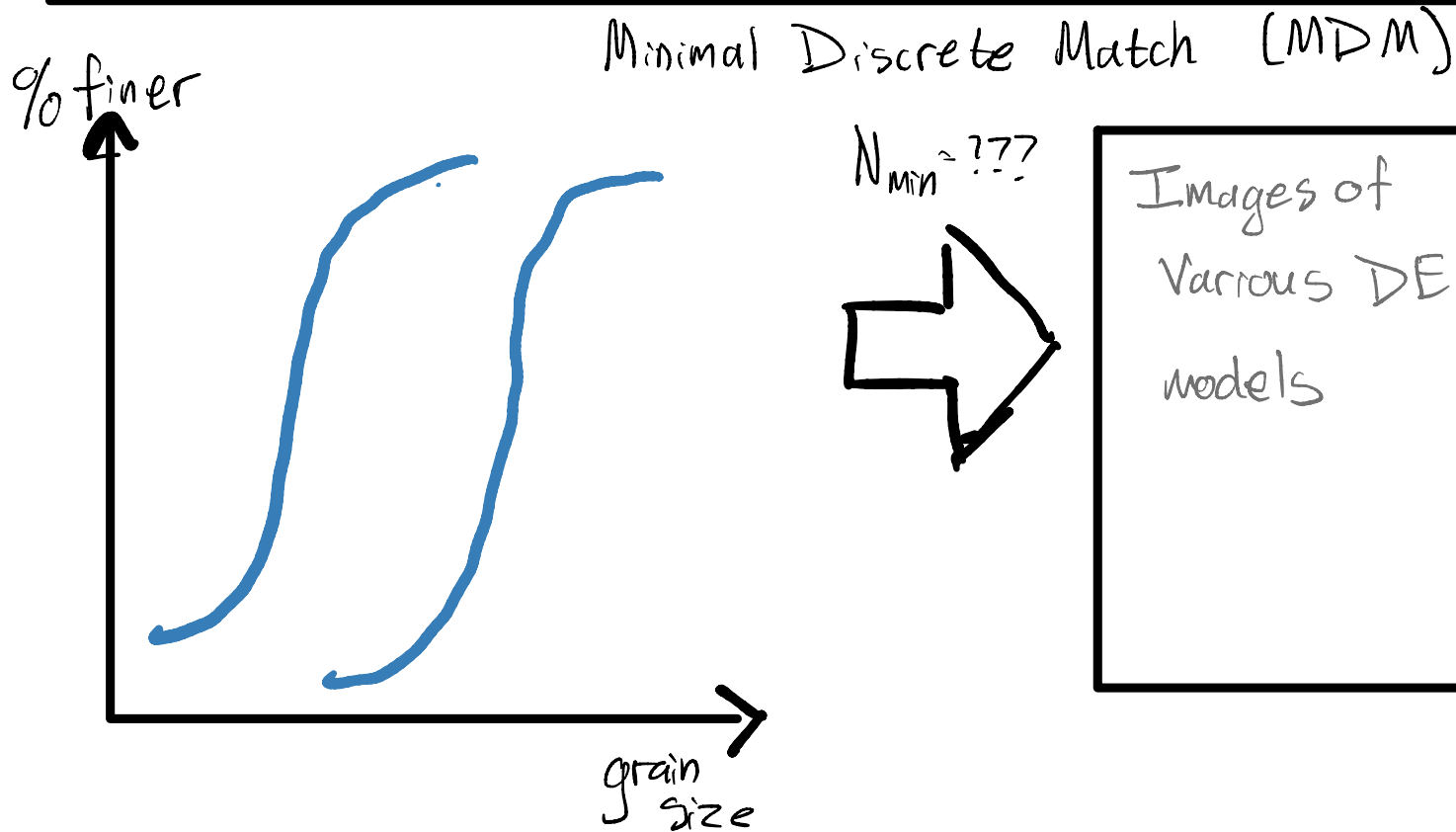
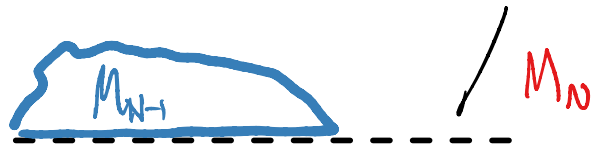
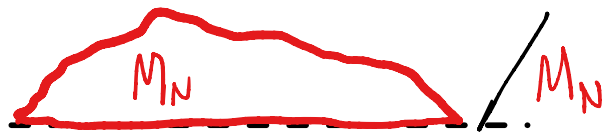


What is the smallest number of discrete particles needed to match a given grain size distribution?



Images of
Various DEM
models

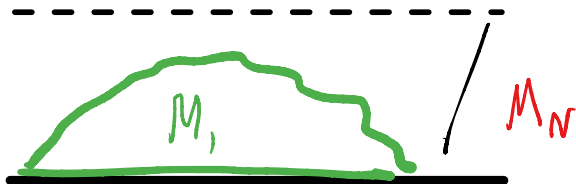
Total Mass Ratios



•

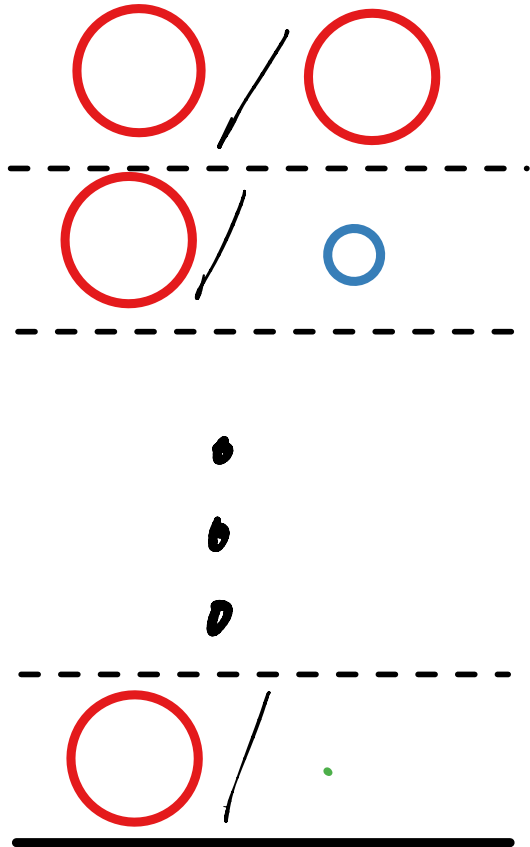
•

•



$$\left\{ \begin{array}{c} 1 \\ \mathbb{R} \\ \vdots \\ 0 \\ \mathbb{R} \end{array} \right\} = \underline{\underline{\mathbb{I}}}$$

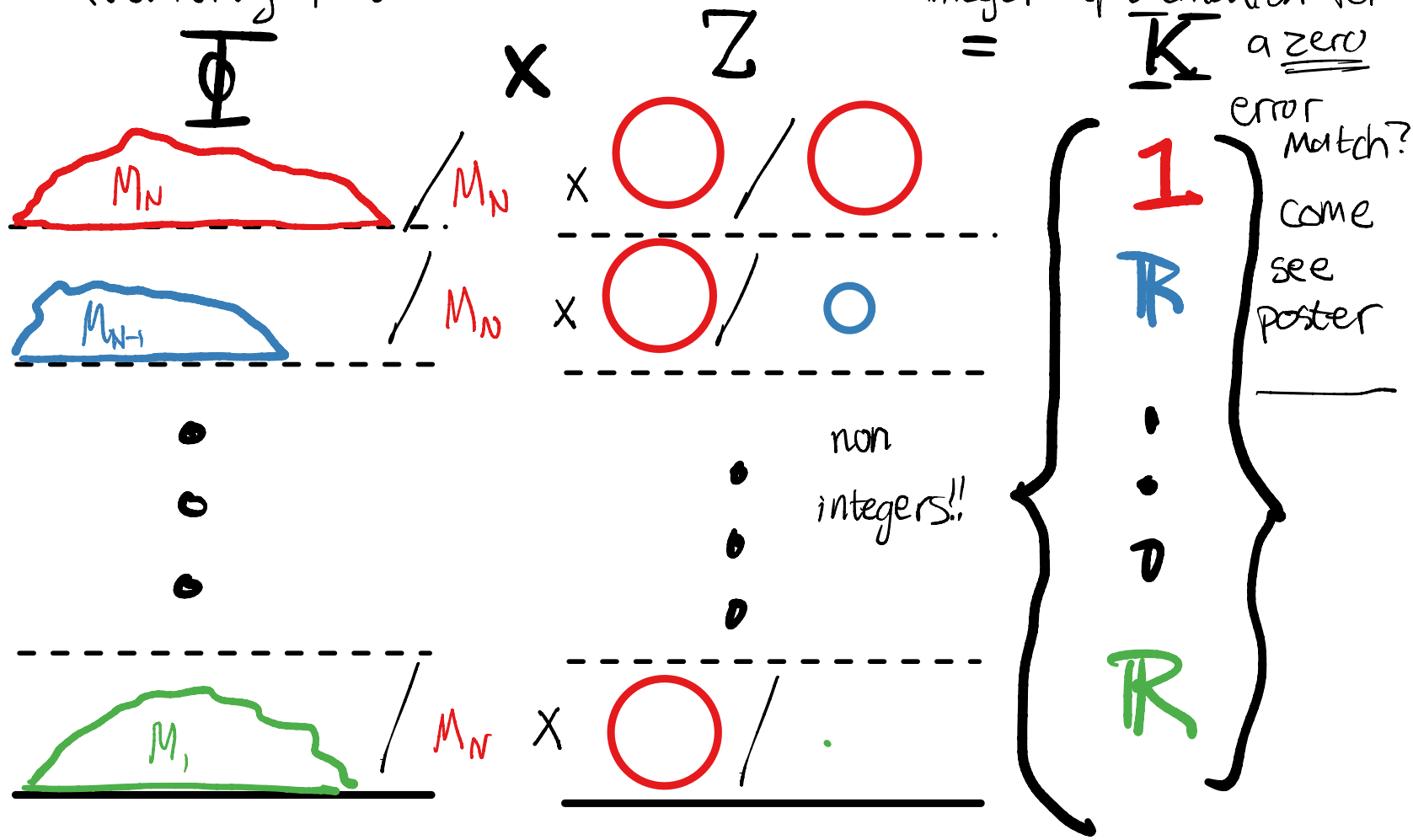
Particle "Volume" ratios



$$\left\{ \begin{array}{c} 1 \\ \mathbb{R} > 1 \\ \vdots \\ 0 \\ \mathbb{R} > 1 \end{array} \right\} = Z$$

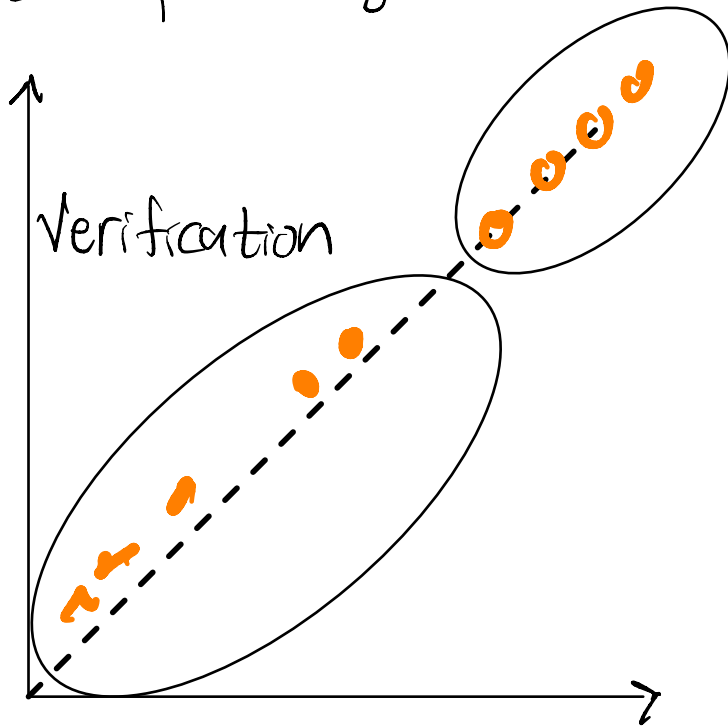
Quantity ratios

How to find minimum possible integer representation for



Verification and
Example usage

$$MDM = N_{\min}(GSD)$$



Experimental design ✓

Any grain size dist. ✓

Arbitrary shapes ✓

Mathematically rigorous ✓

$GSD \Rightarrow$ Computational cost ✓

Representative volume ✗